January 29, 2026: Discussion on XPCS of NBH SSEs

Quick recap

The team conducted a detailed analysis of XPCS data to study particle and grain sizes in crystalline materials, examining how different sample preparation methods affect these properties and exploring scattering patterns at various spatial scales. They observed collective ionic dynamics in cubic materials at nanoscale levels, with Emilio presenting a model that recreates data images using harmonic oscillators to demonstrate correlated motion in the glassy state. The group discussed their findings on ion transport dynamics and anion motion, while planning to publish their current results and explore further studies to understand the underlying physics of borohydride materials.

Summary

Review of State-of-the-Art Understanding

James gave a recap of three regimes of ion transport thought to occur in NBH electrolytes, spanning from a highly ordered, low conductivity phase at low temperature to a more disordered phase at intermediate temperature and eventually a high mobility superionic phase at elevated temperature. The ion transport in these regimes was ascribed to solid-like, glass-like, and liquid-like conduction, and the disordered glassy phase is classified by a high degree of cation-anion correlations and concerted ion motions.

XPCS Data Analysis Techniques

The team discussed XPCS data analysis, focusing on particle and grain sizes, with James explaining that SEM shows particles while XPCS examines grain sizes in highly crystalline materials. They analyzed how different sample preparation methods, such as slow evaporation versus ball milling, affect particle sizes and dynamics. The group examined scattering patterns and masks used to analyze data at different spatial scales, noting that the APS upgrade enabled them to see finer details that would be missed at older facilities. Notably, going to larger dq off the central Bragg peak enabled insight into smaller coherent volumes, and this level of contrast is unachievable at previous lightsources.

Ionic Dynamics in Cubic Materials

The group discussed their findings on collective ionic dynamics in cubic materials at nanoscale levels, observing temperature-dependent behavior and stripes/checkerboard patterns that indicate correlated motion and hint at phenomena such as jamming, avalanching, and memory effects. James and Oleg explored how smaller length scales don't significantly affect dynamics once collective motion is established, suggesting intrinsic behavior rather than length-scale dependence. They agreed to work with Shyue Ping Ong on modeling calculations. The discussion concluded with confirmation that their observations are sample-derived rather than instrumental artifacts, as evidenced by temperature-dependent Bragg Peak changes and Q-dependence.

Harmonic Oscillator Model for Glasses

Emilio presented a model that recreates data images using a mixture of two harmonic oscillators, one with twice the frequency of the other. He explained that this model demonstrates collective ionic dynamics in the glassy state, which is necessary for recovering correlations over long periods. The next steps involve adding an exponential decay term to the model and introducing an "avalanche" effect to study the timescale of memory in the system.

Harmonic Ion Transport Dynamics

Oleg and James discussed the dynamics of ion transport observed in XPCS experiments, where they found evidence of fast, periodic ion movements that could be described as a harmonic oscillator. They noted that these dynamics were faster than the measurement time of half a second, suggesting that the ions move quickly but then slow down in a periodic fashion. James mentioned a jump relaxation model for ion transport, but Shirley emphasized the need to focus on the macroscopic behavior rather than microscopic explanations. They also discussed the possibility of different phases, such as crystalline and amorphous domains, influencing the observed ion dynamics.

Anion Motion and Data Interpretation

The team discussed the anion motion in a material, focusing on the QENS data that shows fast anion libration behavior. They debated the interpretation of this motion, with James suggesting it could be orthogonal libration preferentially allowing sodium conduct in certain planes, while Shirley expressed doubt about the directionality of the motion based on the high symmetry of the cage structure. They also discussed the connection between local probe data and macroscopic structures, emphasizing the need for consistency in interpreting data.

Borohydride Dynamics Research Update

The team discussed their findings on the collective motion and relaxation processes in borohydride materials, focusing on the correlation between atomic-scale dynamics and macroscopic behavior. They agreed to move forward with publishing their current results, which include extracting a critical time constant for the system, while acknowledging the need for further studies to understand the underlying physics. James and Erik will work on refining the figures and storyline for the paper, aiming to complete a draft within the next month. The team plans to meet with Shyue Ping to discuss how to scale the atomic-scale physics to the observed macroscopic behavior. They also considered exploring alternative experimental approaches, such as using amorphous materials or thin films, to better understand the ion dynamics with operando approaches.

Next steps

- James: Prepare the supplement info about the crystal NBH versus ball-milled NBH, including morphology, including XRD, electrochemical, etc., for the publication. Acquire synchrotron XRD if possible.
- Shirley: Reach out to Shyue Ping to request input and calculations on scaling atomic-scale physics to 30-100 nm scale for NBH, and coordinate a meeting with him (targeting February 9 or 11, 4pm San Diego time).
- James, Erik, and Emilio: Refine and complete the first draft of figures for the NBH publication within the next month and circulate for feedback.
- James, Erik, and Emilio: Continue to develop and nail down the storyline and conclusions for the NBH paper, especially regarding the microscopic origins and scaling of the observed dynamics, especially before the February 9/11 meeting with Shyue Ping.
- James and Shen: Design cells (half-cells) for operando XPCS experiments with sodium metal and NBH thin films.
- James: Work with Eric and Emilio to further develop the storyline